

Gridlock

: 4/1/21

1. Lexical Analyzer Design

* TRANSITION TABLE :-

State	input	Next State
S	[\t]	S
S	[A-Z a-z]	ID
S	[0-9]	NUM
S	' + '	PLUS (accept)
S	' - '	HYPHEN
S	' : '	COLON (accept)
S	' \n '	NEWLINE (accept)
S	' = '	EQ1
S	other	DEAD (error)
ID	[A-Za-z 0-9 .]	ID
ID	other	ID (accept, retract)
NUM	[0-9]	NUM
NUM	other	NUM (accept, retract)
HYPHEN	' > '	ARROW (accept)
HYPHEN	other	MINUS (accept, retract)
EQ1	' = '	EQ (accept)
EQ1	other	DEAD (error)

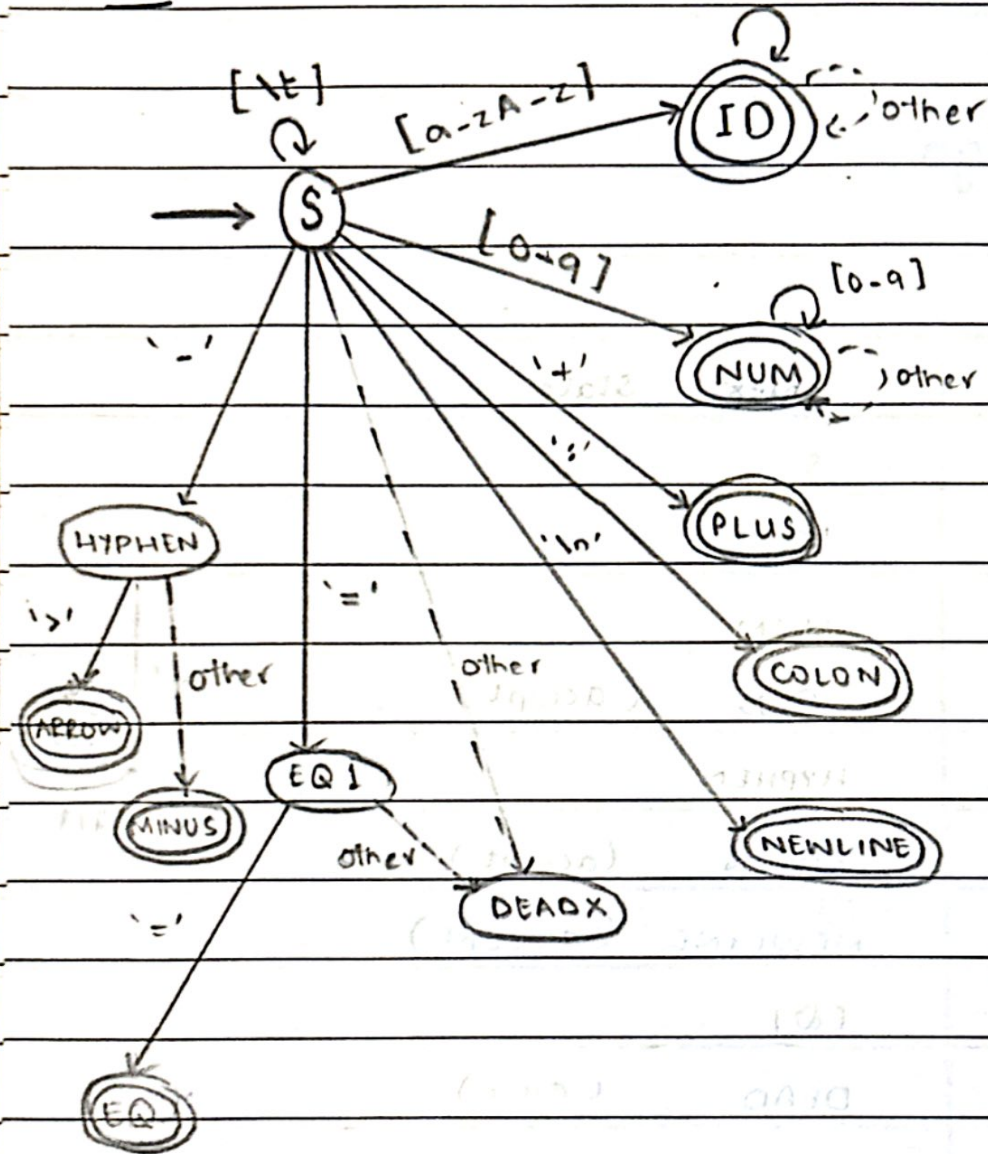
* "retract" => unread last char (put it back into input buffer)

• NOTE :- Keywords (grid, player, enemy, wall, goal, etc) are all lexed as ID tokens and classified by parser.

Date:_____

* DFA:

[A-20-20-9], other



* dashed arrow $\Rightarrow$ retract

(give back last char
to input)

Date: _____

2. Parser Design:

* grammar:

program $::=$ statement *

statement $::=$ grid-stmt | entity-stmt | score-stmt | rule-block | blank-line |
comment-line.

grid-stmt $::=$ "grid" grid-mode NUMBER NUMBER

grid-mode $::=$ "random" | ϵ

entity-stmt $::=$ entity-kind "at" NUMBER NUMBER entity-extra?

entity-kind $::=$ "player" | "enemy" | "wall" | "goal"

entity-extra $::=$ "moves" ID

score-stmt $::=$ "score" sign NUMBER "on" event

sign $::=$ "+" | "-" | ϵ

event $::=$ ID (ID)*

rule-block $::=$ "rule" ":" NEWLINE INDENT rule-line*

rule-line $::=$ "if" ID "=" ID "->" rule-action

rule-action $::=$ "win" | "lose"

blank-line $::=$ ϵ

comment-line $::=$ "+" *

* input program 1 :

grid 10 10

player at 0 0

enemy at 5 5 moves random

wall at 3 3

wall at 3 4

goal at 9 9

Score +10 on reach goal

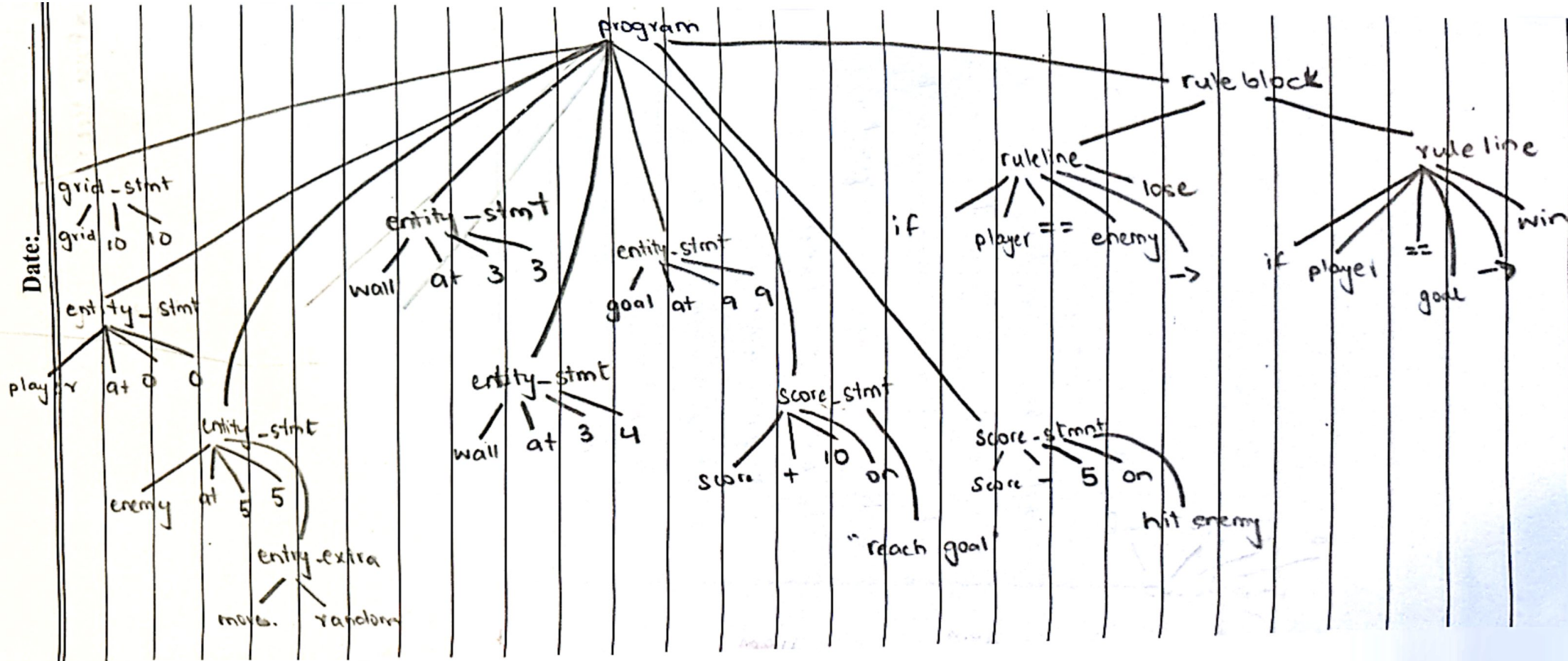
Score -5 on hit enemy

rule :

if player == enemy -> lose

if player == goal -> win

Date: _____



*input program 2:

grid random 8 8

player at 0 0

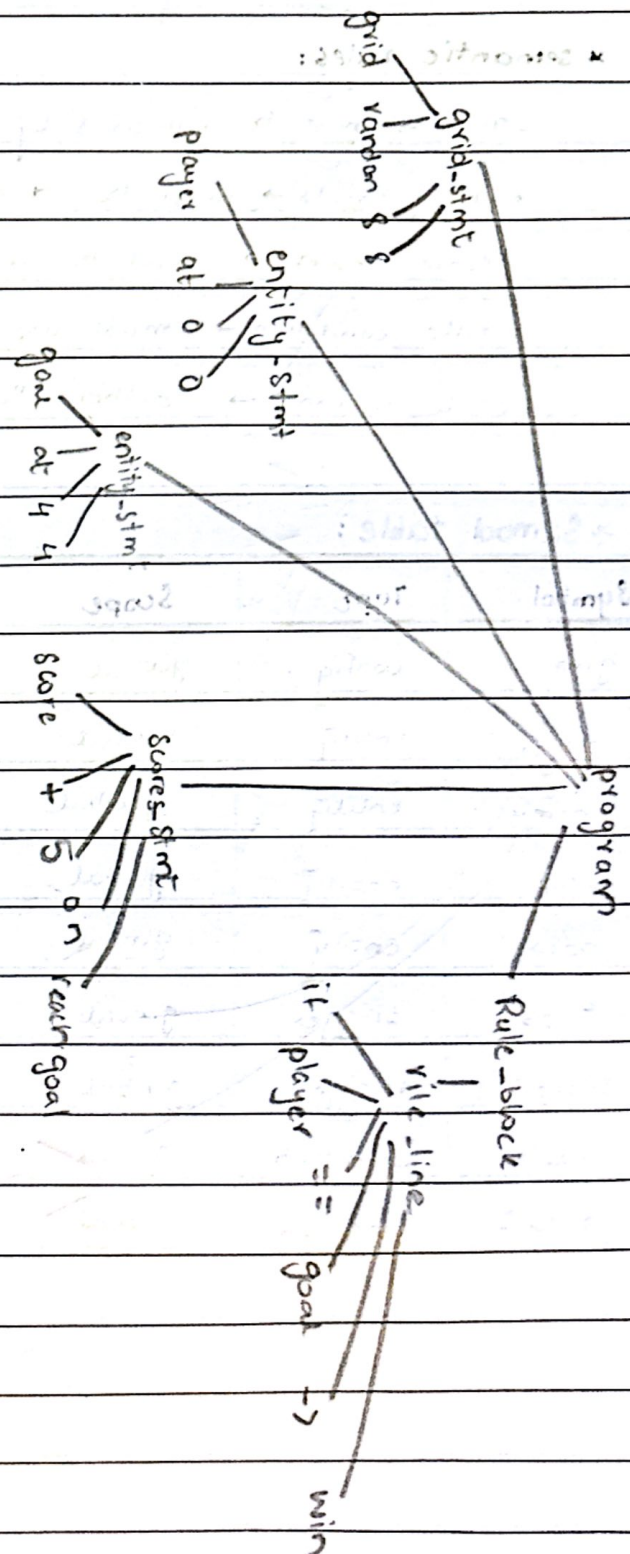
goal at 4 4

score + 5 on reach goal

rule:

if player == goal $\rightarrow$ win

* Parse tree:



3. Semantic Analyzer Design:

* semantic rules:

- grid → must be followed by exactly 2 NUMBERS
- entity coords → must be NUMBERS
- score amount → must be a NUMBER literal
- rule condition → must use ==
- rule action → either "win" or "lose".

* Symbol table:

Symbol	Type	Scope	Attributes / Values
grid	config.	global	width = 10, height = 10
player	entity	global	pos (0,0), moves = None
enemy	entity	global	pos (5,5), moves = random.
wall	entity	global	pos (3,4), moves = None
goal	entity	global	pos (9,9), moves = None
score 1	trigger	global	sign = +, amount = 10, event = "reach goal"
score 2	trigger	global	sign = -, amount = 5, event = "hit enemy"
rule 1	condition	global	lhs = player, op = "==", rhs = enemy, action = lose
rule 2	condition	global	lhs = player, op = "==", rhs = goal, action = win